

NECCDC 2026 — Runbook 3 of 3

Methodology: Finding DC01 on the Unknown Subnet

DC01 is paired with Gitea and Semaphore on the ChefOps core network.

Known: DC02 = 10.100.6.64. Unknown: DC01 subnet and IP. Covers every discovery method with what-ifs for each failure scenario.

Network Context

Based on the competition topology, DC01 lives on the ChefOps core network alongside Gitea, Semaphore, Grafana, Keycloak, Wordpress, and Teleport. This is a separate subnet from your 10.100.6.x network, connected through the dual pfSense firewalls. You are working from DC02 at 10.100.6.64.

What You Know	Value
DC02 IP	10.100.6.64
POS IP	10.100.6.37
Your subnet	10.100.6.x (oceancrest branch)
DC01 subnet	Unknown — ChefOps core side of pfSense
DC01 neighbors	Gitea, Semaphore, Grafana, Keycloak, Wordpress, Teleport

Method 1 — Ask AD Directly (Fastest)

Your DC02 already has a replication relationship with DC01. AD knows where every DC is. This should be your first attempt — it takes 10 seconds and requires no scanning.

Commands to Run on DC02

```
# Ask AD for all DCs in the forest
Get-ADDomainController -Filter * |
    Select Name, IPv4Address, Site, Domain, IsGlobalCatalog

# Ask the forest directly
Get-ADForest | Select -ExpandProperty GlobalCatalogs
Get-ADForest | Select Domains, Sites

# Get DCs per domain
Get-ADForest | Select -ExpandProperty Domains | ForEach-Object {
    Write-Host "`n=== Domain: $_ ==="
    Get-ADDomainController -Filter * -Server $_ |
        Select Name, IPv4Address, Domain
}
```

What If: This returns only DC02

■ If only DC02 appears, replication between DC01 and DC02 may be broken, or DC01 is offline. Proceed to Method 2.

```
# Check replication status
repadmin /showrepl
repadmin /replsummary

# Check if DC01 is even known to AD
nltest /dclist:chefops.local
```

What If: nltest shows DC01 hostname but no IP

```
# Resolve the hostname via DNS
Resolve-DnsName DC01
Resolve-DnsName DC01.chefops.local
nslookup DC01.chefops.local
```

Method 2 — DNS SRV Record Lookup

Every DC automatically registers SRV records in DNS when it comes online. These records contain the DC hostname and IP and are queryable by anyone on the network.

Query DC SRV Records

```
# DCs register under _ldap._tcp.dc._msdcs
nslookup -type=SRV _ldap._tcp.dc._msdcs.chefops.local

# Also check Kerberos SRV records
nslookup -type=SRV _kerberos._tcp.dc._msdcs.chefops.local

# PowerShell DNS query
Resolve-DnsName -Name '_ldap._tcp.dc._msdcs.chefops.local' -Type SRV

# Query all DC-related DNS records
Get-DnsServerResourceRecord -ZoneName 'chefops.local' `
  -RRType A -ErrorAction SilentlyContinue |
  Where-Object {$_.HostName -like 'DC*' -or $_.HostName -like '*dc*'} |
  Select HostName, RecordData
```

What If: DNS returns hostname but ping fails

```
# DNS record exists but machine may be on different subnet
# Extract the IP from the DNS record and test specific ports
$ip = (Resolve-DnsName 'DC01.chefops.local').IPAddress
Test-NetConnection -ComputerName $ip -Port 389
Test-NetConnection -ComputerName $ip -Port 3389
```

What If: DNS query times out completely

■ DNS may be restricted across the pfSense firewall. Port 53 may be blocked between subnets. Move to Method 3.

Method 3 — Discover the Subnet Via Routing

Before you can scan for DC01, you need to know what subnet it's on. Your routing table and network configuration will tell you what subnets are reachable from DC02.

Find the Other Subnet

```
# Full routing table
route print

# PowerShell routing table
Get-NetRoute | Select DestinationPrefix, NextHop, RouteMetric |
  Where-Object {$_.DestinationPrefix -ne '0.0.0.0/0' -and
    $_.DestinationPrefix -ne '127.0.0.0/8'} |
  Sort DestinationPrefix

# Check network adapters for multiple subnets
Get-NetIPAddress | Select IPAddress, PrefixLength, InterfaceAlias

# Check ARP table for recently seen hosts on other subnets
arp -a
```

What If: Routing table shows no other subnets

■ pfSense may not be advertising routes to DC02. Try pinging common gateway IPs to discover what is reachable.

```
# Test common gateway addresses
@('10.100.1.1', '10.100.0.1', '10.0.0.1', '192.168.1.1') | ForEach-Object {
  $r = Test-Connection -ComputerName $_ -Count 1 -ErrorAction SilentlyContinue
  if ($r) { Write-Host "Gateway reachable: $_" }
}

# Try traceroute to see what hops exist
tracert 10.100.6.64 # your own DC - see the path
tracert 8.8.8.8    # follow the path out to find gateway IPs
```

Method 4 — Targeted Port Scan for DC01

Once you know the subnet, scan for the specific ports that only a DC runs. Port 389 (LDAP) and port 88 (Kerberos) together on one IP means DC. Replace X.X with the subnet octets you discovered.

Scan for DC — Port 389 + 88

```
# Scan entire /24 subnet for LDAP (port 389)
# Replace 10.X.X with your discovered subnet
1..254 | ForEach-Object {
    $ip = "10.X.X.$_"
    $ldap = Test-NetConnection -ComputerName $ip -Port 389 `
        -WarningAction SilentlyContinue -InformationLevel Quiet
    if ($ldap) { Write-Host "LDAP open at: $ip" }
}
```

```
# Verify with Kerberos port 88 on candidates found above
$candidates = @('10.X.X.Y') # IPs from LDAP scan
foreach ($ip in $candidates) {
    $ldap = Test-NetConnection -ComputerName $ip -Port 389 `
        -WarningAction SilentlyContinue -InformationLevel Quiet
    $kerb = Test-NetConnection -ComputerName $ip -Port 88 `
        -WarningAction SilentlyContinue -InformationLevel Quiet
    $rdp = Test-NetConnection -ComputerName $ip -Port 3389 `
        -WarningAction SilentlyContinue -InformationLevel Quiet
    Write-Host "$ip - LDAP:$ldap Kerberos:$kerb RDP:$rdp"
}
```

What If: Port 389 scan finds nothing

```
# DC01 may be blocking LDAP from external subnets
# Try Kerberos (88) instead - harder to block
1..254 | ForEach-Object {
    $ip = "10.X.X.$_"
    $r = Test-NetConnection -ComputerName $ip -Port 88 `
        -WarningAction SilentlyContinue -InformationLevel Quiet
    if ($r) { Write-Host "Kerberos at: $ip" }
}
```

What If: Scan is too slow

```
# Run scan in parallel using jobs
$jobs = 1..254 | ForEach-Object {
    $ip = "10.X.X.$_"
    Start-Job -ScriptBlock {
        param($ip)
        $r = Test-NetConnection -ComputerName $ip -Port 389 `
            -WarningAction SilentlyContinue -InformationLevel Quiet
        if ($r) { $ip }
    } -ArgumentList $ip
}
$jobs | Wait-Job | Receive-Job
$jobs | Remove-Job
```

Method 5 — Find Subnet via Gitea or Semaphore

You know DC01 is paired with Gitea and Semaphore. Find either of those services and you know the subnet. Both run on well-known ports.

Scan for Gitea (Port 3000)

```
# Gitea default port is 3000
# Try common subnets first before full scan
$subnets = @('10.100.1', '10.100.2', '10.100.0', '10.0.0', '192.168.1')
foreach ($subnet in $subnets) {
    1..254 | ForEach-Object {
        $ip = "$subnet.$_"
        $r = Test-NetConnection -ComputerName $ip -Port 3000 `
            -WarningAction SilentlyContinue -InformationLevel Quiet
        if ($r) { Write-Host "Gitea found at: $ip" }
    }
}
```

Scan for Grafana (Port 3000 or 80)

```
# Grafana also runs on 3000 by default, or 80 if proxied
1..254 | ForEach-Object {
    $ip = "10.X.X.$_"
    $r80 = Test-NetConnection $ip -Port 80 -WarningAction SilentlyContinue -InformationLevel Quiet
    $r3000 = Test-NetConnection $ip -Port 3000 -WarningAction SilentlyContinue -InformationLevel Quiet
    if ($r80 -or $r3000) { Write-Host "Web service at: $ip (80:$r80 3000:$r3000)" }
}
```

Scan for Keycloak (Port 8080 or 8443)

```
1..254 | ForEach-Object {
    $ip = "10.X.X.$_"
    $r = Test-NetConnection -ComputerName $ip -Port 8080 `
        -WarningAction SilentlyContinue -InformationLevel Quiet
    if ($r) { Write-Host "Keycloak possibly at: $ip" }
}
```

✓ Once any of these services responds, you have the subnet. DC01 will be another IP on that same subnet. Run the port 389+88 scan on just that subnet to find it.

Method 6 — Query pfSense

pfSense sits between your subnet and DC01's subnet. If your team has access to the pfSense admin console, it will show you the exact routing table and connected interfaces — giving you the subnet directly.

pfSense Admin Console

- Navigate to the pfSense web UI — typically at the gateway IP on your subnet
- Login with competition credentials
- Go to: Diagnostics > Routes — shows all known subnets
- Go to: Interfaces — shows all connected subnets with IP ranges
- Go to: Diagnostics > ARP Table — shows all recently active IPs

If You Cannot Access pfSense UI — Try From DC02

```
# Ping pfSense gateway
Test-Connection -ComputerName -Count 2

# Try pfSense web UI port
Test-NetConnection -ComputerName -Port 443
Test-NetConnection -ComputerName -Port 80

# Check your default gateway - that IS pfSense
(Get-NetIPConfiguration).IPv4DefaultGateway.NextHop
```

Decision Tree — Which Method to Use

Situation	Method to Use
Have not tried anything yet	Method 1 — Ask AD (Get-ADDomainController)
AD returns only DC02	Method 2 — DNS SRV records
DNS times out or fails	Method 3 — Check routing table for subnet
Have subnet, need DC IP	Method 4 — Port 389+88 scan
Subnet unknown, scan too slow	Method 5 — Find Gitea/Grafana first
Team has pfSense access	Method 6 — pfSense ARP/routes table
All methods fail	Ask @BlackTeam for DC01 hostname/IP (0 points if competent question)

Once DC01 IP Is Found — Confirm It Is a DC

```
# Confirm it responds on all DC ports
$dc1 = ''
@(53,88,135,389,445,636,3268,3389) | ForEach-Object {
    $r = Test-NetConnection -ComputerName $dc1 -Port $_ `
        -WarningAction SilentlyContinue
    Write-Host "Port $_ : $($r.TcpTestSucceeded)"
}

# Confirm it is actually running AD
Get-ADDomainController -Server $dc1

# Get its full details
Get-ADDomainController -Identity $dc1 |
    Select Name, IPv4Address, Domain, Site, IsGlobalCatalog, OperationMasterRoles
```

After Confirming DC01 — Immediately Connect and Harden

```
# RDP to DC01
mstsc /v:

# Alternatively via PowerShell remoting
Enter-PSSession -ComputerName -Credential (Get-Credential)

# Once in: run Runbook 1 from Phase 0
```

■ DC01 has had zero hardening. Run the full Runbook 1 hardening sequence immediately. Every minute it is unhardened is a window for Red Team to re-establish persistence before you can clean it.

Black Team Ticket Template (Last Resort)

```
@BlackTeam
We are attempting to locate DC01 on the ChefOps core network.
We have tried:
- Get-ADDomainController: returns only DC02 (10.100.6.64)
- DNS SRV lookup: [result]
```

- Route table: shows only 10.100.6.0/24
- Port 389 scan on [subnet]: no response

Can you confirm the hostname or IP of DC01?

(Competent question - prior effort documented above)